

Skills Summary

One-Sided, Infinite, and End-Behavior Limits

Use this reference while completing your worksheet or when studying.

Three different questions wear the same notation. Read the arrow first: a superscript $\pm$ means *pick a side*, a finite target with a zero denominator means *find a sign*, and $x \rightarrow \pm\infty$ means *compare degrees*.

1. One-sided limits

Rule: $x \rightarrow c^-$ uses only values just below c ; $x \rightarrow c^+$ only values just above. The two-sided limit exists exactly when both exist and agree.

This is how absolute values and piecewise rules get resolved: the side tells you which branch, or which sign, is in force.

Example: $\lim_{x \rightarrow 3^-} \frac{|x-3|}{x-3}$

Step 1. An absolute value cannot be simplified without knowing a sign, and the $-$ superscript supplies it.

Step 2. For $x < 3$, $x - 3 < 0$, so $|x - 3| = -(x - 3)$ and the quotient is -1 for every such x .
 -1

Try it: $\lim_{x \rightarrow 4^+} \frac{|x-4|}{x-4}$

check: 1

2. Infinite limits at a vertical asymptote

Rule: nonzero numerator over a denominator going to 0 is $\pm\infty$. Get the sign from the denominator on the side you are on:

$$\frac{(+)}{(0^+)} \rightarrow \infty, \quad \frac{(+)}{(0^-)} \rightarrow -\infty$$

Test a nearby value such as $c + 0.01$ if the sign is not obvious. Never report a number.

Example: $\lim_{x \rightarrow 2^+} \frac{5}{x-2}$

Step 1. The numerator is a nonzero constant while the denominator collapses to zero, so the answer is infinite and only the sign is in doubt.

Step 2. Just above 2, $x - 2$ is a small positive number, and 5 over a small positive number is large and positive.

∞

Try it: $\lim_{x \rightarrow 1^-} \frac{3}{x-1}$

check: $-\infty$

3. End behavior as x goes to infinity

Rule for a rational function: compare degrees.

- top degree $<$ bottom degree: limit is 0
- degrees equal: limit is the ratio of leading coefficients
- top degree $>$ bottom degree: no finite limit ($\pm\infty$)

The safe method underneath all three: divide every term by the highest power of x in the denominator. Inside a square root that means dividing by x^2 , and $\sqrt{x^2} = -x$ when $x \rightarrow -\infty$.

Example: $\lim_{x \rightarrow \infty} \frac{5x^2 - x}{2x^2 + 3}$

Step 1. Both parts have degree 2, so the tails grow at the same rate and the limit is the ratio of the leading coefficients.

Step 2. Dividing through by x^2 gives $\frac{5 - 1/x}{2 + 3/x^2} \rightarrow \frac{5}{2}$.

$$\frac{5}{2}$$

Try it: $\lim_{x \rightarrow -\infty} \frac{6x^3 + x}{3x^3 - 2}$

check:

4. Making a piecewise function continuous

Rule: f is continuous at $x = c$ when

$$\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = f(c)$$

To find an unknown constant, evaluate each branch at c , set the two expressions equal, and solve. One equation, one unknown.

Example: Find a so that $f(x) = 3x + a$ for $x < 1$ and $f(x) = x^2 + 5$ for $x \geq 1$ is continuous at $x = 1$.

Step 1. Continuity is exactly the statement that the two one-sided limits agree, which turns the question into one equation.

Step 2. Left: $3 + a$. Right: $1 + 5 = 6$. So $3 + a = 6$.

$$a = 3$$

Try it: Find c so that $f(x) = 2x + c$ for $x < 4$ and $f(x) = x^2 - 3$ for $x \geq 4$ is continuous at $x = 4$.

check: